

Stock Risk + Sentiment

Predicting Volatility Risk — and Testing Whether Sentiment Helps

Capstone Project Report

An educational model that predicts a stock's next-week volatility risk from price and news sentiment
— and the honest finding that sentiment didn't help.

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1. Executive Summary

This project predicts whether a stock's coming week will be Low, Medium, or High **volatility** (how much it moves — not which direction) from 12 price and news-sentiment features, using a small neural network. It was built mainly to answer one question honestly: does the sentiment actually add anything?

The model reaches **56% accuracy on three classes** (chance is 33%), driven by the fact that volatility clusters in time. But the headline is the ablation: a **price-only** model scored **57.2%** — slightly better. **The sentiment did not help.**

This is an educational machine-learning project, **not financial advice** and not a recommendation to buy, sell, or hold anything.

2. Introduction & Motivation

Predicting a stock's direction is close to a coin flip; predicting its volatility is more tractable because calm and turbulent periods cluster. The interesting question for a project called '...with sentiment' is whether a news-sentiment signal earns its place alongside price features.

3. Data

Daily prices for 59 stocks came from Yahoo Finance; ticker-level news sentiment (positive/negative/neutral per article) came from a public 2023 Polygon news dataset on Kaggle. From these, 12 features were built per stock-day — trailing returns, rolling volatility, volume, drawdown, and a daily sentiment signal.

No look-ahead: every feature uses only past data, the label is the **next** week's realised volatility, and train/test are split **by time** with a gap, so no test label overlaps the training period.

4. Methodology

A small multilayer perceptron (3,397 parameters) classifies the feature vector into volatility terciles. To test sentiment's value, an identical model was trained on the **same features minus the three sentiment ones**, and the two were compared on the held-out later period.

5. Results

Model (unseen period)	Accuracy
Price + sentiment	56.3%
Price only	57.2%
Chance (3 classes)	33.3%

Volatility is genuinely predictable (well above chance), but sentiment added **-0.9 points** — nothing, within noise. That is an honest and common finding: a daily average of headline sentiment is a coarse, sparse signal, and price-based volatility features already capture most of what's predictable.

Reporting the negative result — instead of quietly dropping the ablation — is the difference between a demo and an experiment.

6. Deployment

Exported to a 20 KB ONNX model for an in-browser demo (pick a real stock-and-week and see the predicted risk vs. what actually happened) and to TensorFlow Lite for an Android app ('Risk Explorer'), both verified to match PyTorch to $\sim 4e-7$.

7. Limitations & Ethical Considerations

- Educational only — not financial advice and not a recommendation.
- Predicts volatility, not direction; it is right only about half the time.
- One year of data and one coarse sentiment signal.

8. Future Work

- A richer sentiment signal: per-headline embeddings rather than a daily average.
- A longer, multi-year history spanning more market regimes.
- Test whether sentiment helps more for direction than for volatility.